Chi-Square

A test that measure how a model compares to actual observed data
or Help us by determining whether these two variables are associated with each other.
Comparing 2 categoricals

For example, Entity 1: People's favorite colors and Entity 2: Their preference for ice cream.

- **Null Hypothesis (H_0):** Favorite color and ice cream preference are ***independent** (no relationship).
- ***Alternative Hypothesis (H_1):*** They are **dependent** (a relationship exists).

Formula For Chi-Square Test

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Symbols are broken down as follows:

- **O_i:** Observed frequency
- **E_i:** Expected frequency

If this value is large enough, we reject H_0, concluding that color preference does influence ice cream choice and vice versa.

	Chocolate	Vanilla	Strawberry
Male	20	15	10
Female	25	20	30

Observed frequency is the table given above.

$$E_{ij} = \frac{(Row\ Total) \times (Column\ Total)}{Grand\ Total}$$

- Male and chocolate: $\frac{45 \times 45}{120} = 16.875$
- Male and Vanilla: $\frac{45 \times 35}{120} = 13.125$

Summarizing,

- **Male**: Chocolate: 16.875, Vanilla: 13.125, Strawberry: 15.0,
- **Female:** Chocolate:12.125, Vanilla: 21.875, Strawberry: 25.0

Step 4: Perform Chi-Square Test

Example

Solved Examples on Chi-Square Test

Example 1: A study investigates the relationship between eye color (blue, brown, green) and hair color (blonde, brunette, Redhead). The following data is collected:

Eye Color	Blonde	Brunette	Redhead	Total
Blue	35	52.5	12.5	100
Brown	28.1	42.1	9.8	80
Green	6.9	10.4	2.7	20

Solution:

Calculate the chi-square value for each cell in the contingency table using the formula

$\chi^2 = (O_i - E_i)^2 / E_i$

For instance, consider someone with brown hair and blue eyes:

$\chi^2 = (15 - 28.1)^2 / 28.1 \approx 6.07.$

To complete the total chi-square statistic, find each cell's chi-squared value and sum them up across all the nine cells in the table.

Degrees of Freedom (df):

df = (number of rows - 1) × (number of columns - 1)

df = (3 - 1) × (3 - 1)

df = 2 × 2 = 4

Finding p-value:

You may reference a chi-square distribution table to get an estimated chi-square stat of (χ^2) using the appropriate degrees of freedom. Look for the closest value and its corresponding p-value since most tables do not show precise numbers.

If your Chi-square value was 20.5, you would observe that the nearest number in the table for df = 4 is 14.88 with a p-value in 0.005; an illustration is.

Interpreting Results:

- *Selecting a level of significance (α = 0.05 is common)or than if the null hypothesis holds, the probability of either rejecting it at all is limited (Type I error).*
- *Compare the alpha value and p-value.*
- *When the p-value is less than the significance level, which in this case is written as p-value < 0.05, we can reject the null hypothesis. There is sufficient evidence to say that hair and eye color are related in one direction according to statistical terms. If the p-value is greater than the significance level it means that we cannot reject the null hypothesis therefore p-value > 0.05.*
- *Based on the data at hand, we cannot say that there is a statistically significant correlation between eye and hair colors.*

Main types of chi-square tests There are **two primary types** of Pearson's chi-square tests that are used most often:

1. **Chi-square goodness-of-fit test**
   - **When to use**: You have **one single categorical variable** and want to see if its observed distribution matches some **expected/theoretical distribution**.
   - **Null hypothesis (H_0)**: The observed frequencies match the expected frequencies (the data fits the expected model).
   - **Common examples**:
     - Is a 6-sided die fair? (Should each face appear ~16.67% of the time?)

    - Do the sales of 4 ice cream flavors match the company's expected market shares (e.g., 40% chocolate, 30% vanilla, 20% strawberry, 10% others)?
    - Does the distribution of car colors in your sample match national percentages?
  - **Data format**: One row of observed counts vs. one row of expected counts (or proportions).
2. **Chi-square test of independence** (also called test of association)
  - **When to use**: You have **two categorical variables** and want to test whether they are **independent** (unrelated) or if there is an association between them.
  - **Null hypothesis (H_0)**: The two variables are independent (no relationship between them).
  - **Common examples**:
    - Is gender related to preferred smartphone brand?
    - Does smoking status (smoker/non-smoker) relate to having lung disease (yes/no)?
    - Is education level associated with voting preference?
    - In marketing: Is region related to product satisfaction rating?
  - **Data format**: A **contingency table** (crosstab) with rows = one variable, columns = the other, and cells = counts.
3. **Chi-square test of homogeneity**
  - **When to use**: You have **multiple independent groups** (populations/samples) and want to test whether they have the **same distribution** for one categorical variable (i.e., proportions are homogeneous across groups).
  - **Null hypothesis (H_0)**: The distributions are the same across all groups.
  - **Key difference from independence test**: In independence, you usually have one sample and two variables measured on each unit. In homogeneity, you have separate samples (groups) and one variable measured on each.
  - **Example**: Do different cities have the same proportion of people who prefer tea vs. coffee? (Separate random samples from each city.)
  - **Mathematically**: The calculation is identical to the test of independence — the distinction is mainly in how you interpret the question and sampling design.

When to use the chi-square test

Use the **chi-square (χ^2) test** when your data is **categorical** (counts, frequencies, or categories like yes/no, colors, gender, blood type, survey choices, etc.) and you want to test whether the observed data significantly differs from what you'd expect under some hypothesis — basically, to check if patterns are real or just due to random chance.

You should use it when:

- Your variables are **categorical** (nominal or ordinal with categories).
- You have **count/frequency data** (not means of continuous data).

- Sample size is large enough → generally, **expected counts ≥ 5** in most (or all) cells of the table.
- You want to test **independence**, **goodness-of-fit**, or **homogeneity** (see types below).
- Data comes from a random sample.

Do NOT use chi-square if:

- Your data is continuous (use t-test, ANOVA, etc.).
- You have very small expected counts (<5 in many cells) → consider Fisher's exact test instead.
- You're comparing means of numerical data.